

SIMATS ENGINEERING

CO4 – AT1 - Article Review- Low

COURSE NAME: Advance Wireless Communication Systems /

5G / 6G Communication Systems

COURSE CODE: ECA5607

COURSE OUTCOME COVERED

CO: 5 - Develop 5G-enabled edge computing and IoT solutions for smart city, healthcare, industrial automation, and intelligent communication applications. (BL6)

Assessment Tool Weightage: 25%

Short description about assessment: Students review recent technical articles to assess critical thinking and awareness of trends.

Dr. TAMIL SELVI M

QUESTIONS:

Total Marks:

Duration: 90 Minutes

1. Multi-access Edge Computing (MEC) Architectures for Low-Latency 5G Applications (BL5)
2. Role of MEC in Ultra-Reliable Low-Latency Communications (URLLC) (BL5)
3. Cloud Radio Access Network (Cloud RAN): Architecture, Benefits, and Challenges (BL5)
4. **Virtualization and Resource Management in Cloud RAN (BL5)**
5. Edge AI for Intelligent 5G Network Optimization (BL5)
6. Artificial Intelligence-Based Resource Allocation in Edge Computing (BL5)
7. Integration of MEC and Artificial Intelligence for Next-Generation Networks (BL5)
8. Security and Privacy Challenges in Edge Computing for 5G (BL5)
9. Dynamic Spectrum Sharing Techniques for Efficient Spectrum Utilization (BL5)
10. AI-Driven Dynamic Spectrum Sharing in 5G and Beyond Networks (BL5)
11. Satellite Backhaul Solutions for Rural and Remote 5G Connectivity (BL5)
12. Low Earth Orbit (LEO) Satellite Networks for 5G and 6G Communications (BL5)
13. Comparative Analysis of LEO, MEO, and GEO Satellites for Wireless Communication (BL4)
14. Integration of Terrestrial and Non-Terrestrial Networks (NTN) in 6G (BL5)
15. Edge Computing over Satellite Networks for Future Wireless Systems (BL5)

16. NB-IoT Architecture and Emerging Industrial Applications (BL5)
17. LTE-M Technology for Massive Machine-Type Communications (BL5)
18. Comparative Study of NB-IoT and LTE-M for IoT Deployments (BL4)
19. Power-Efficient Communication Techniques for IoT Devices in 5G Networks (BL5)
20. Battery Life Optimization Strategies for Massive IoT Devices (BL5)
21. Wearable Healthcare Devices Enabled by 5G Communication (BL5)
22. Smart Sensors for Intelligent IoT Applications in 5G Networks (BL5)
23. Energy-Efficient Communication Protocols for Large-Scale IoT Systems (BL5)
24. Intelligent Traffic Management Using 5G and Edge Computing (BL5)
25. AI-Enabled Smart Surveillance Systems over 5G Networks (BL5)
26. Smart Energy Monitoring and Management Using IoT and 5G (BL5)
27. Digital Twin Technology for Smart City Infrastructure (BL5)
28. Edge AI Applications for Sustainable Smart Cities (BL5)
29. Industrial IoT Applications in Smart Manufacturing (BL5)
30. Predictive Maintenance Using Artificial Intelligence and 5G Networks (BL5)
31. Robotic Automation in Industry 4.0 Using 5G Communication (BL5)
32. Private 5G Networks for Smart Factory Deployments (BL5)
33. Edge Computing for Real-Time Industrial Automation (BL5)
34. Remote Robotic Surgery Enabled by 5G Communication Networks (BL5)
35. AI-Driven Medical Diagnosis Using 5G and Edge Computing (BL5)
36. Intelligent Remote Patient Monitoring Using Wearable IoT Devices (BL5)
37. Secure Healthcare IoT Systems in 5G Networks: Challenges and Solutions (BL5)
38. Smart Irrigation Systems Using IoT and 5G Technologies (BL5)
39. Precision Agriculture for Crop Monitoring and Yield Prediction Using AI and 5G (BL5)
40. Intelligent Livestock Tracking and Smart Farming Using 5G IoT Networks (BL5)

Code of Conduct:

I B. PRAKASH (192412025) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A handwritten signature in black ink, appearing to read 'B. Prakash', is written over a horizontal line.

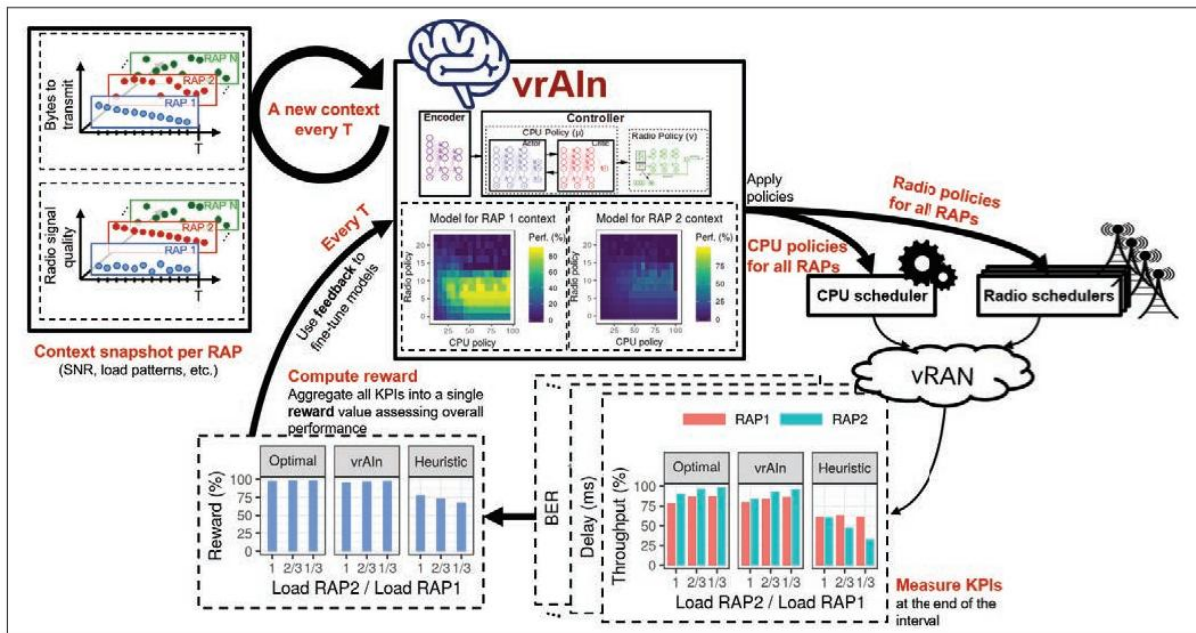
Virtualization and Resource Management in Cloud RAN

Abstract:

The O-RAN Alliance represents a global initiative to establish new levels of openness in next-generation virtualized radio access networks (vRANs) [cite: 1]. Initially launched by five major mobile carriers, the alliance has grown to include over 160 companies, demonstrating exceptional international collaboration to define novel technical standards [cite: 1]. By incorporating AI/ML-based innovations, O-RAN presents a disruptive potential for multi-vendor deployments.

Introduction:

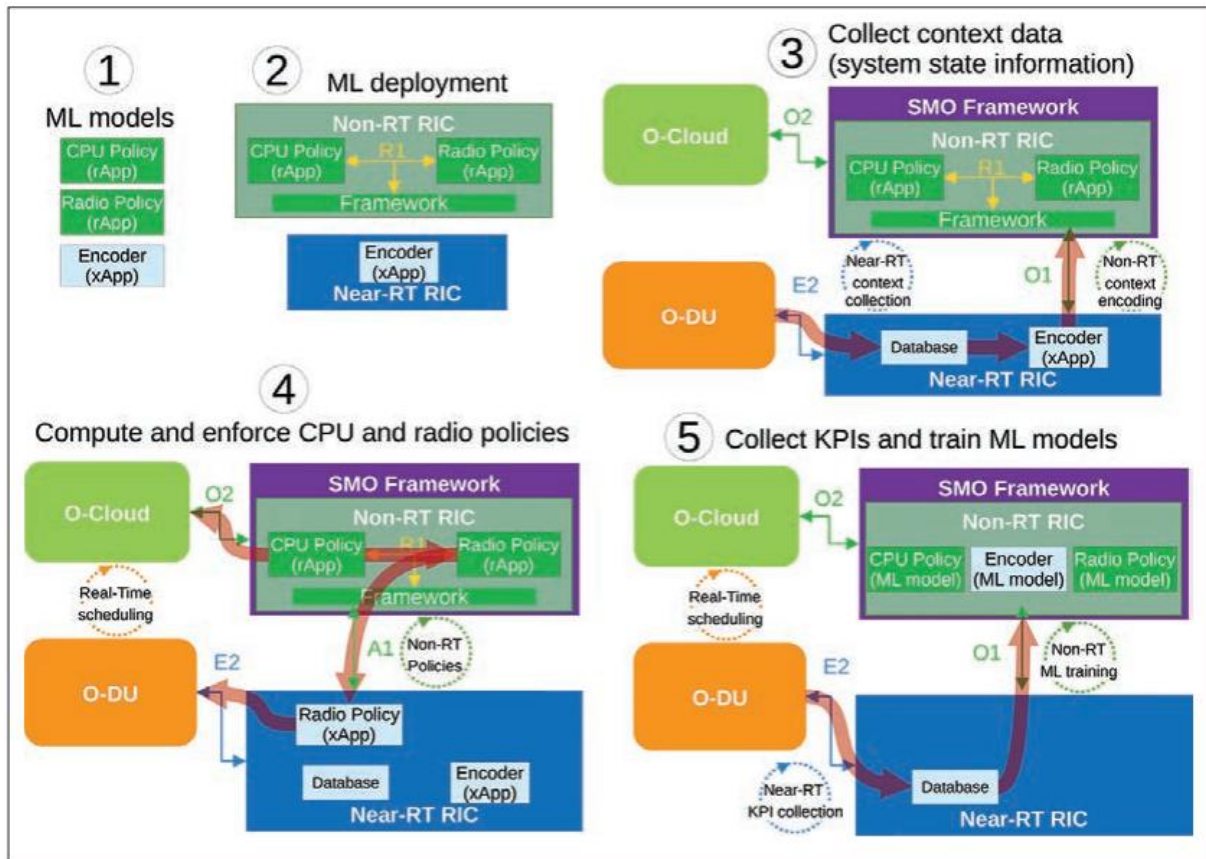
Virtualization of radio access networks offers significant operational and capital expenditure (OPEX/CAPEX) savings, along with high flexibility [cite: 1]. O-RAN is leading the effort to define next-generation vRANs to break vendor lock-in and open a market traditionally dominated by a small set of players [cite: 1]. If successful, this initiative is expected to unleash unprecedented innovation in the RAN space by lowering the entry barrier for new competitors.



AI-aided approach to joint computing/radio resource orchestration

O-RAN Architecture:

The O-RAN architecture introduces several critical functional components [cite: 1]. The non-real-time (non-RT) radio intelligent controller (RIC) is hosted by the service management and orchestration (SMO) framework, while the near-RT RIC can be co-located with 3GPP gNB functions (such as O-CU and O-DU) or fully decoupled [cite: 1]. Additionally, the O-Cloud provides an O-RAN compliant platform utilizing hardware acceleration when necessary.



Integration of an AI-aided vRAN resource orchestrator into O-RAN architecture.

Deployment Options:

O-RAN's approach fosters different degrees of openness, ranging from O-RAN-compliant physical network functions to a fully decoupled fabric of commercial off-the-shelf servers [cite: 1]. This flexibility allows operators to decide whether to co-locate O-DU and O-RU components depending on context and capacity [cite: 1]. Different scenarios divide tasks between regional clouds, edge clouds, and proprietary or O-Cloud cell sites.

Conclusion

O-RAN has achieved exceptional momentum, yielding over 40 technical specifications and 1.3 million lines of open-source code in just two years [cite: 1]. While it faces significant challenges to reach the economies of scale needed to compete with established vendors, its shared O-Cloud approach and AI-aided orchestration hold massive disruptive potential for the telecommunications industry.

RUBRICS FOR CALCULATION:

Article Review					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand